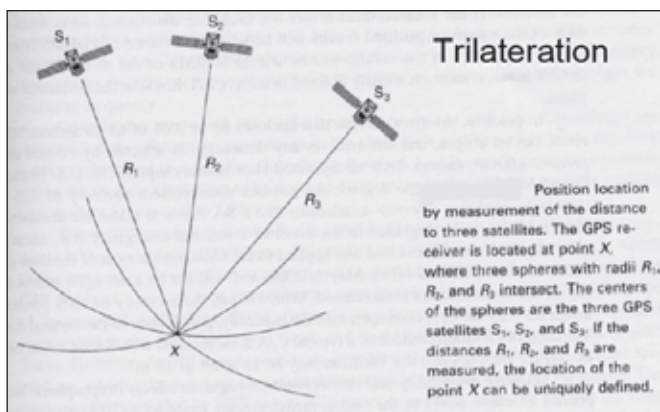




The GPS receiver calculates its position by analysing the signals from three satellites

satellites each sends out a sphere of signals that all travel with the same speed in all directions (since they are electromagnetic waves, they travel at the speed of light, i.e. around 3,00,000 km per second or 1,86,000 miles per second). It is the intersection of at least three of these spheres that enables the calculation of coordinates. With just three spheres, the object can be at any altitude above the calculated coordinates with respect to the Earth. However, if a fourth satellite is also used, then the exact altitude can also be calculated.

To recap, receivers also have a clock that is synchronised with the satellites, thus, it can calculate the amount of time taken for the signal from each of the satellites to reach it. Thus, using basic distance conversion formulae, it can calculate its exact distance from each of the satellites. Then, using mathematics, and with the prior knowledge of the location of the satellites, it is possible for it to locate the coordinates in a matter of seconds.

However, there are many possible causes of errors, although these are usually minor.

Satellite geometry can affect the quality of GPS signals and accuracy of receiver trilateration.

Dilution of Precision (DOP) reflects each satellite's position relative to the other satellites being accessed by a receiver.